PRE-PRESS TECHNOLOGIES

GRAPHIC DESIGN & MULTIMEDIA PROGRAM

COURSE CODE: GDM5310

ACADEMIC YEAR: 2026/2027

LEVEL: 3

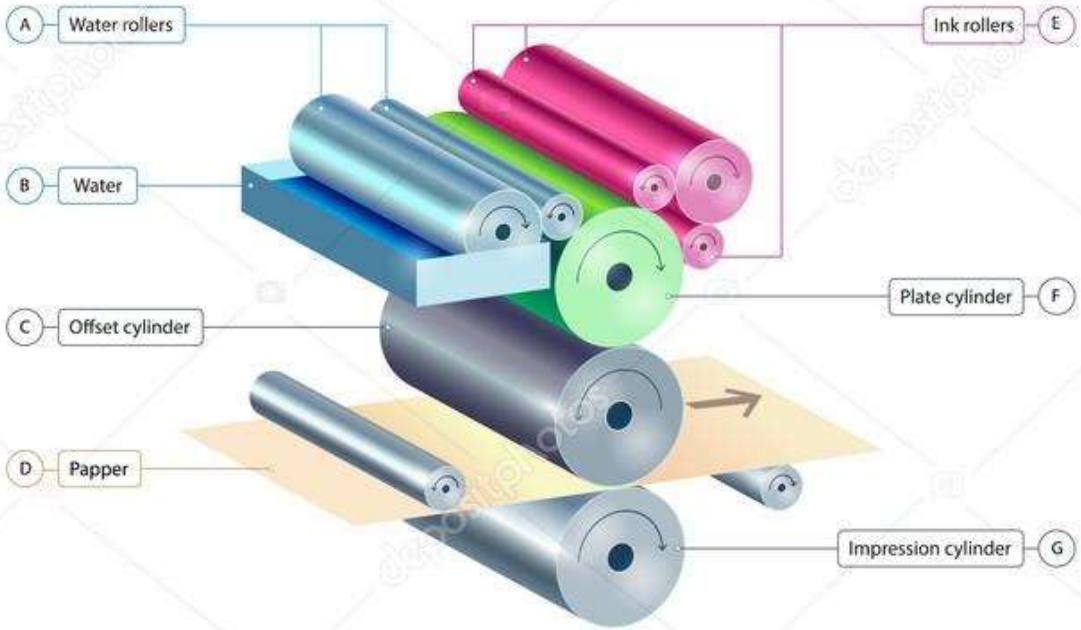
LECTURE 2

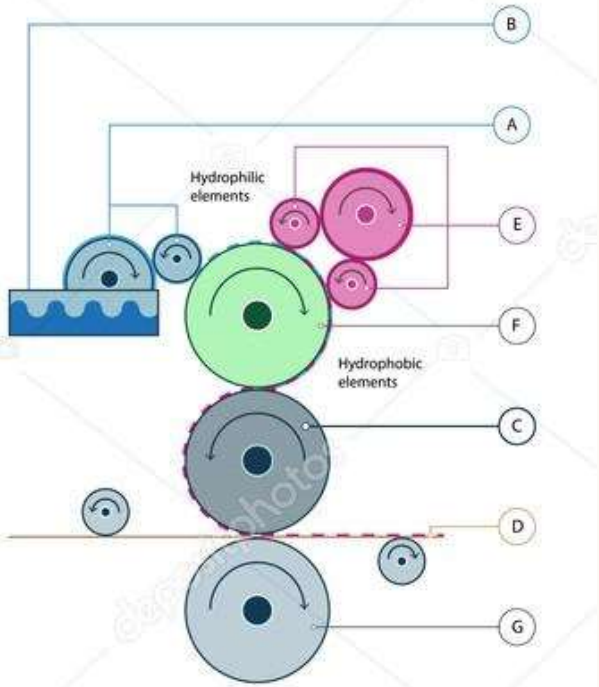
BY: DR/ RANIA MOHAMED

AGENDA

- Revision
- Lithographic plate types
- Lithographic plate preparation
- Safety in Lithography Surface Preparation Departments

Offset printing





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LITHOGRAPHIC PRINTING PREPARATION

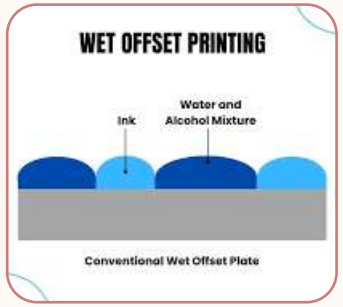


File Preparation

Plate Preparation

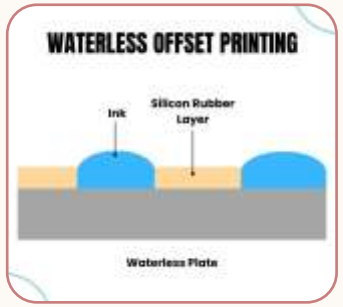


PLATE PREPARATION



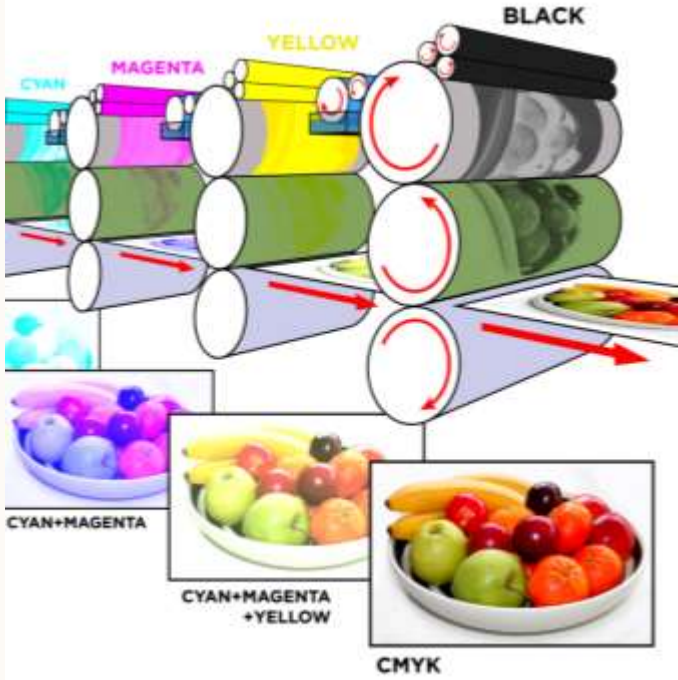
Conventional Printing Plates

- Use a water softener solution during printing

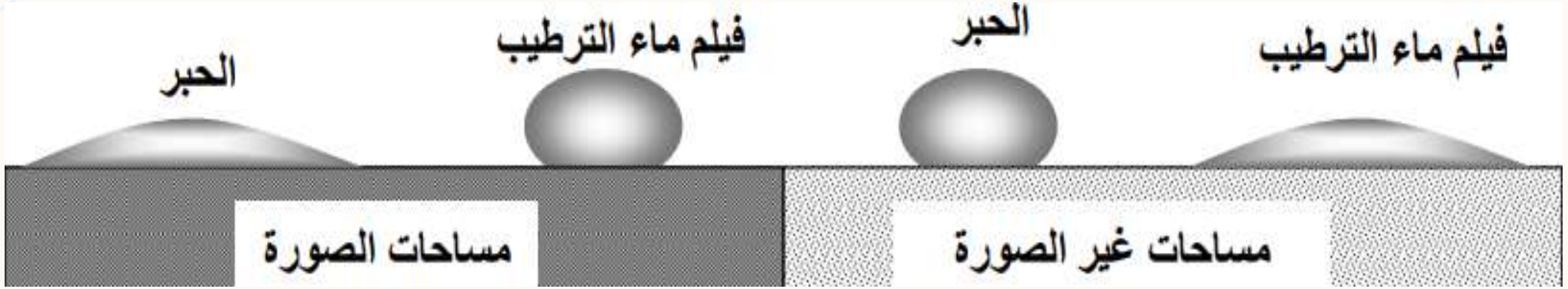


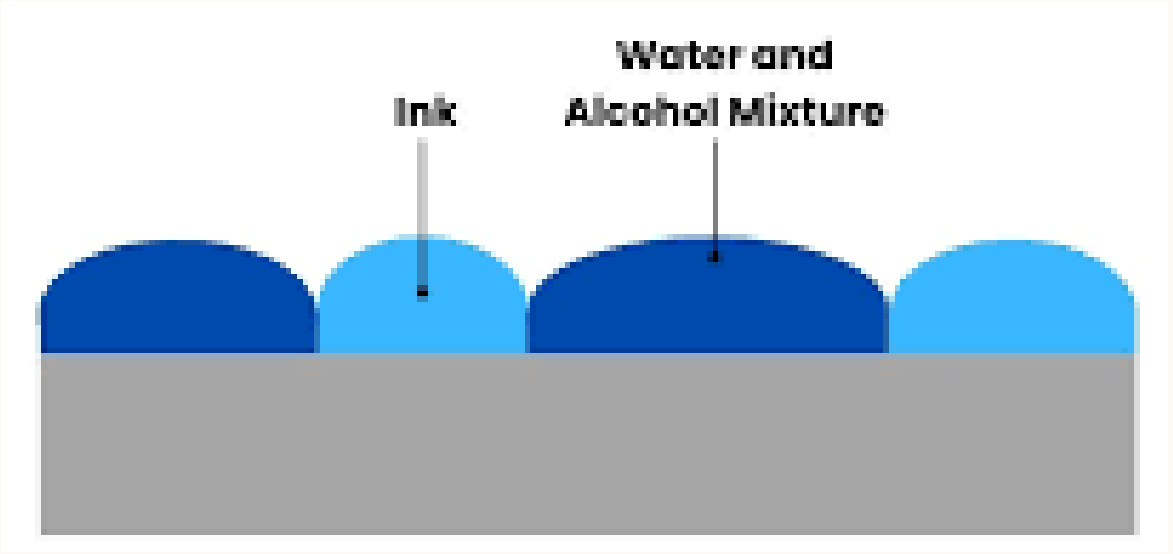
Waterless printing plates

- Do not use a water softener solution during printing



CONVENTIONAL PRINTING PLATES

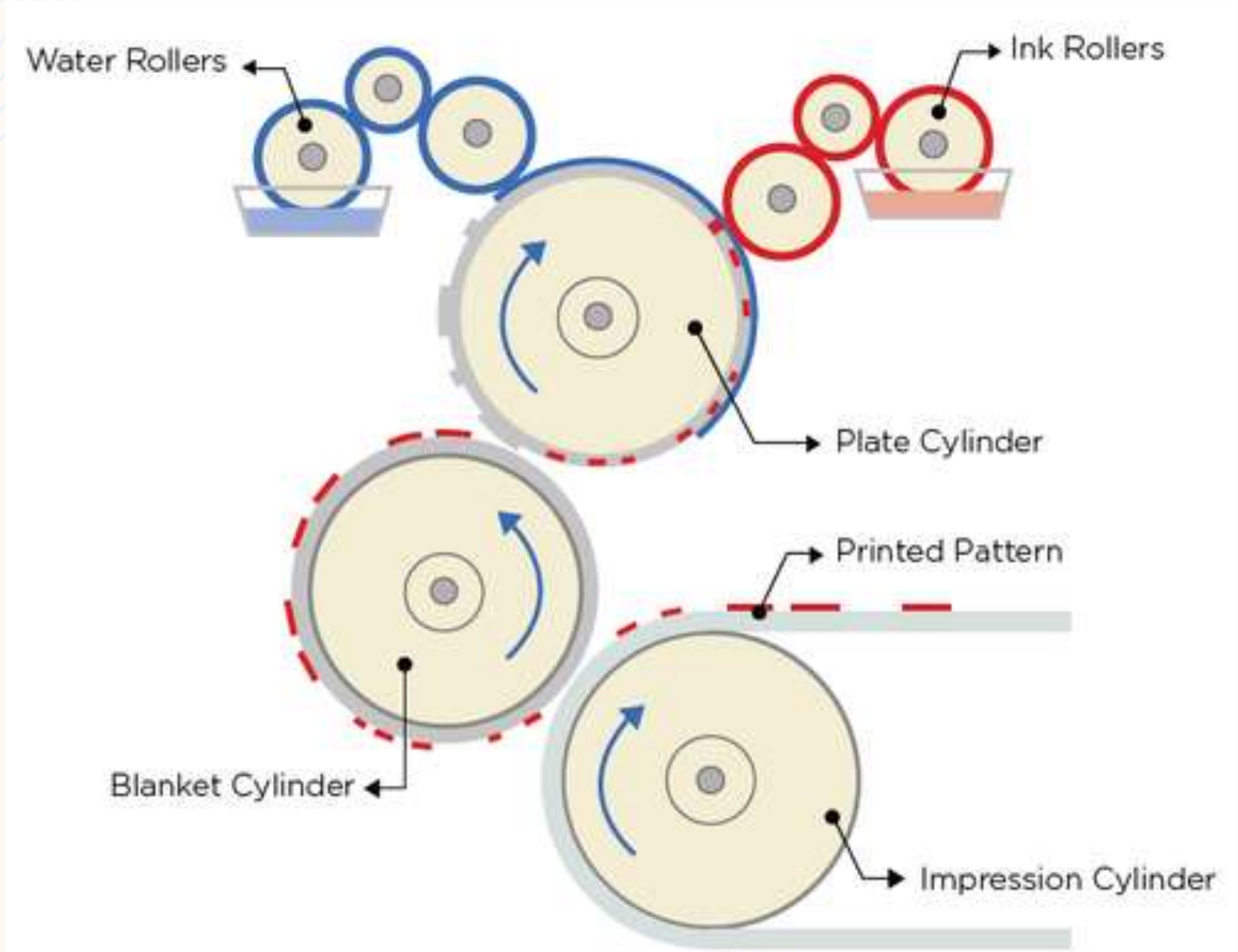




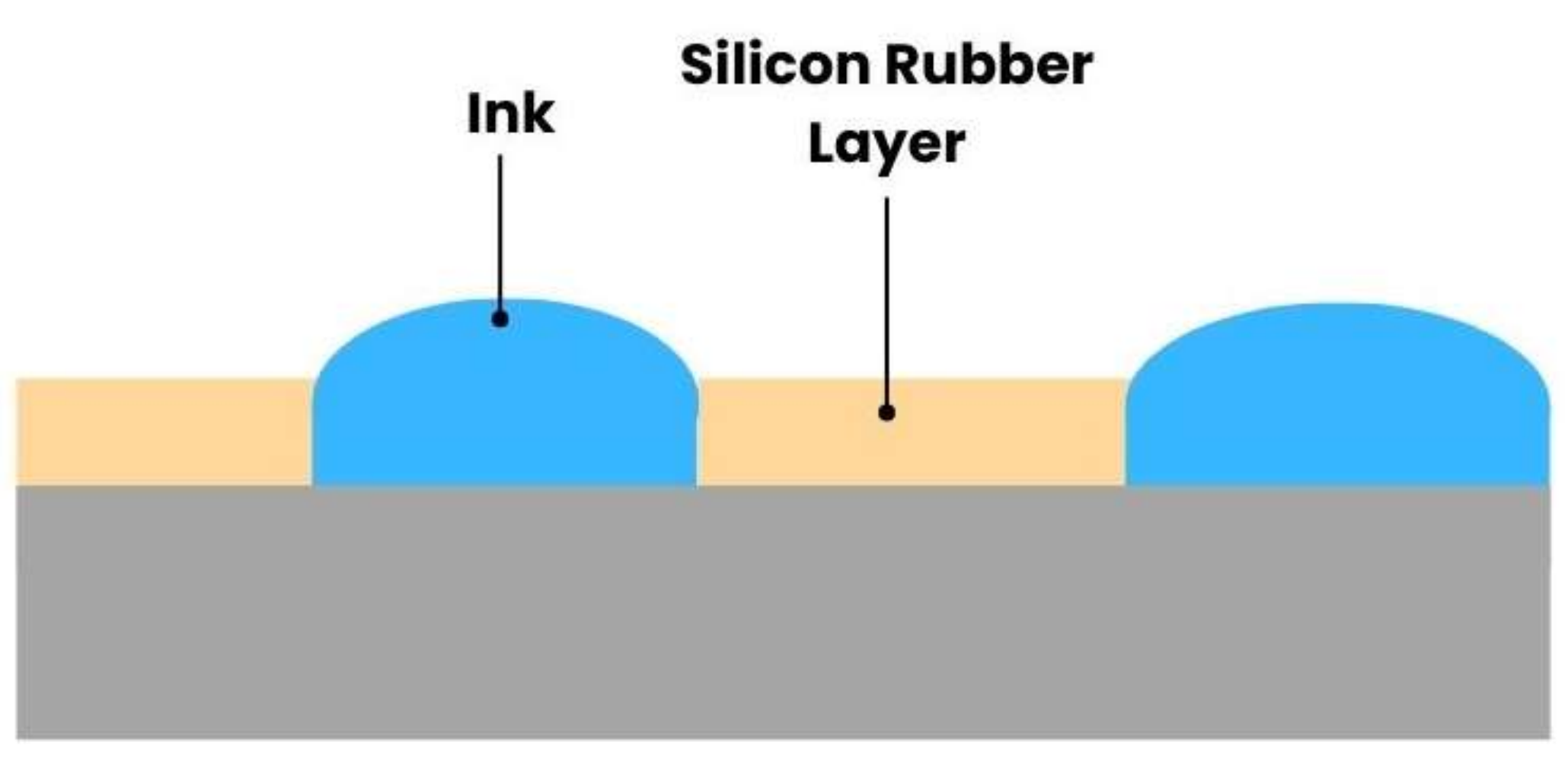
Conventional „Wet“ Printing

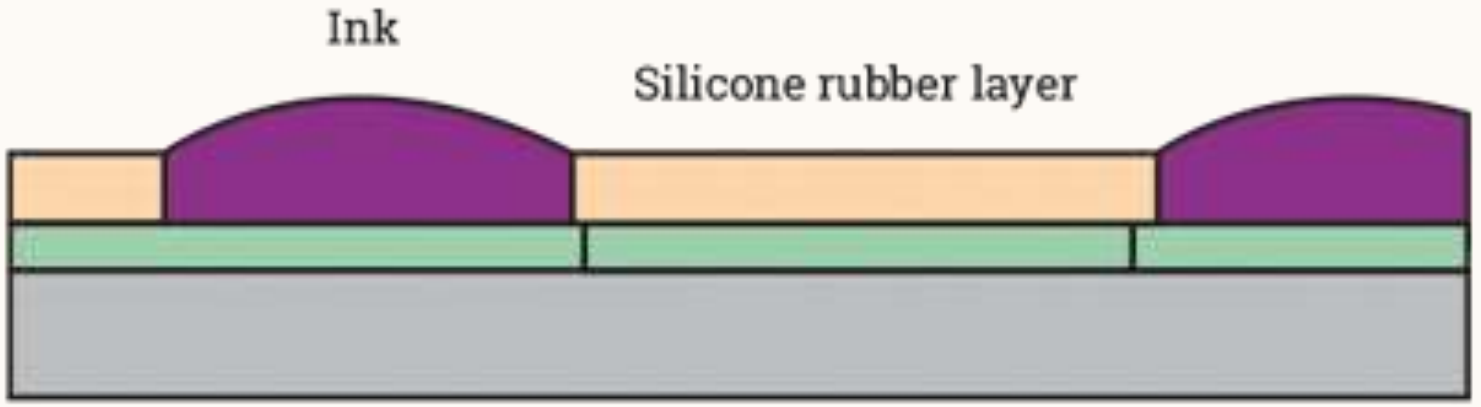


Dampening solution (shown in blue, below left) is used to repel ink (in magenta). This results in tainted water being discharged from the press.

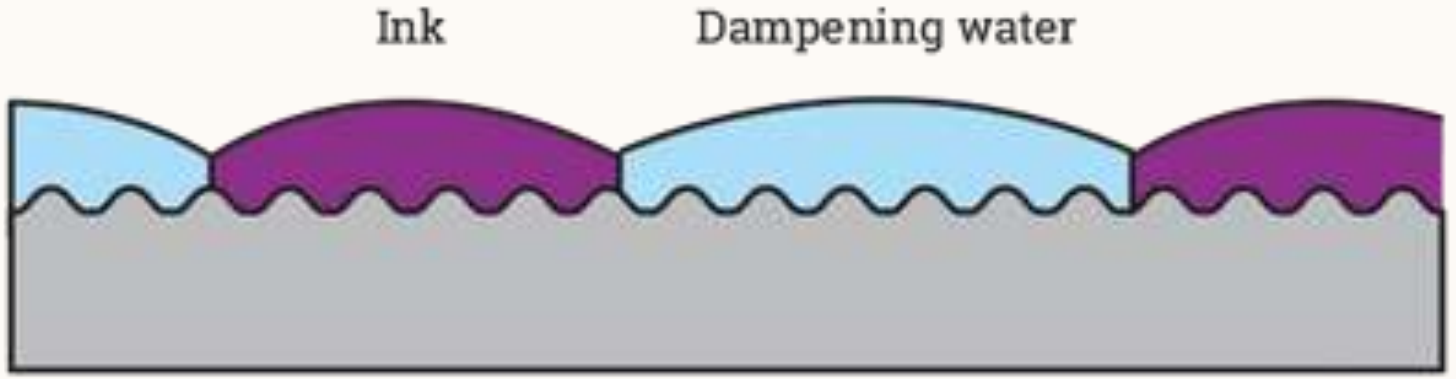

Water Rollers
Ink Rollers
Plate Cylinder
Printed Pattern
Blanket Cylinder
Impression Cylinder

WATERLESS PRINTING PLATES





Waterless plate



Conventional wet offset plate

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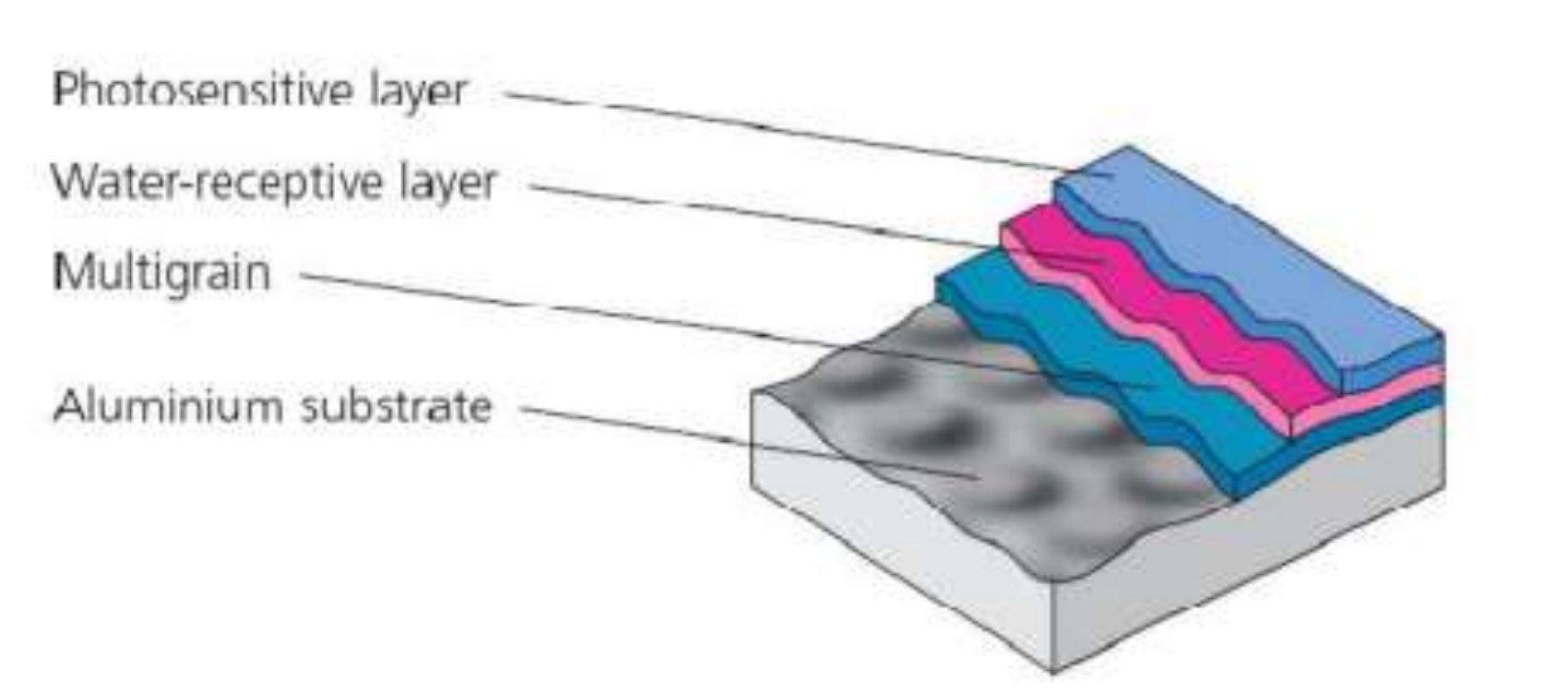
Classification of printing plates according to the type of substrate

- Paper plates
- Plastic plates
- Metal plates

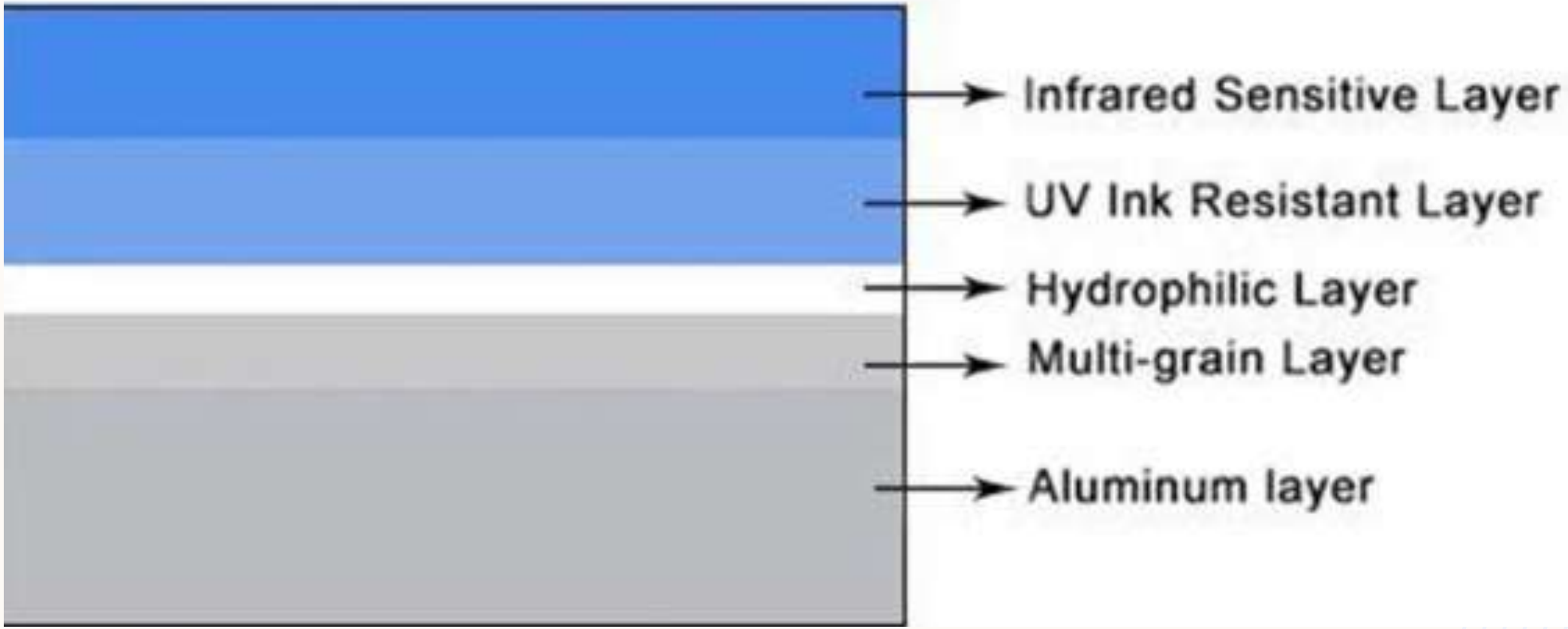
A - METAL PLATES

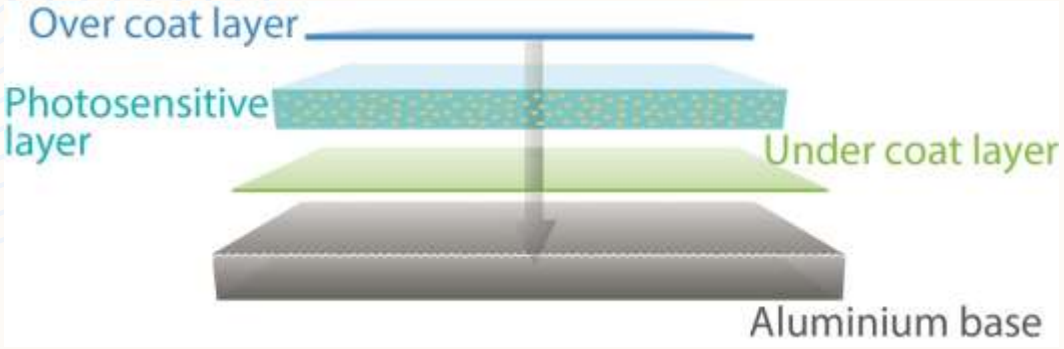
- Metal plates are used for printing large-volume works because they have a longer lifespan compared to other types. Aluminum is used as a support, similar to traditional analog plates, after being granulated. This allows the plate to retain a thin film of wetting solution in the non-printing areas. This keeps these areas free of ink during printing, while the printing areas become ink-receptive.

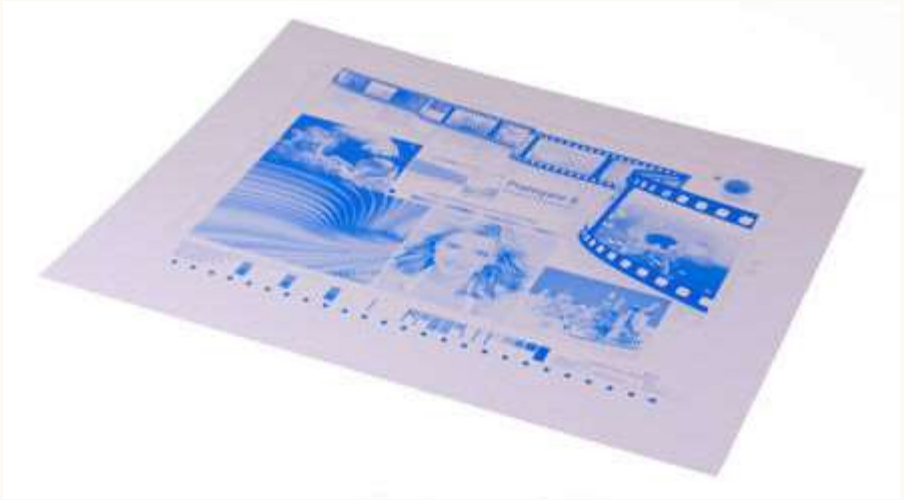
- The plates are coated with various types of photosensitive coatings, including silver halide and photopolymers, depending on the plate type and the processors used.
- Metal plates are used for printing from 100,000 to 2 million or more.

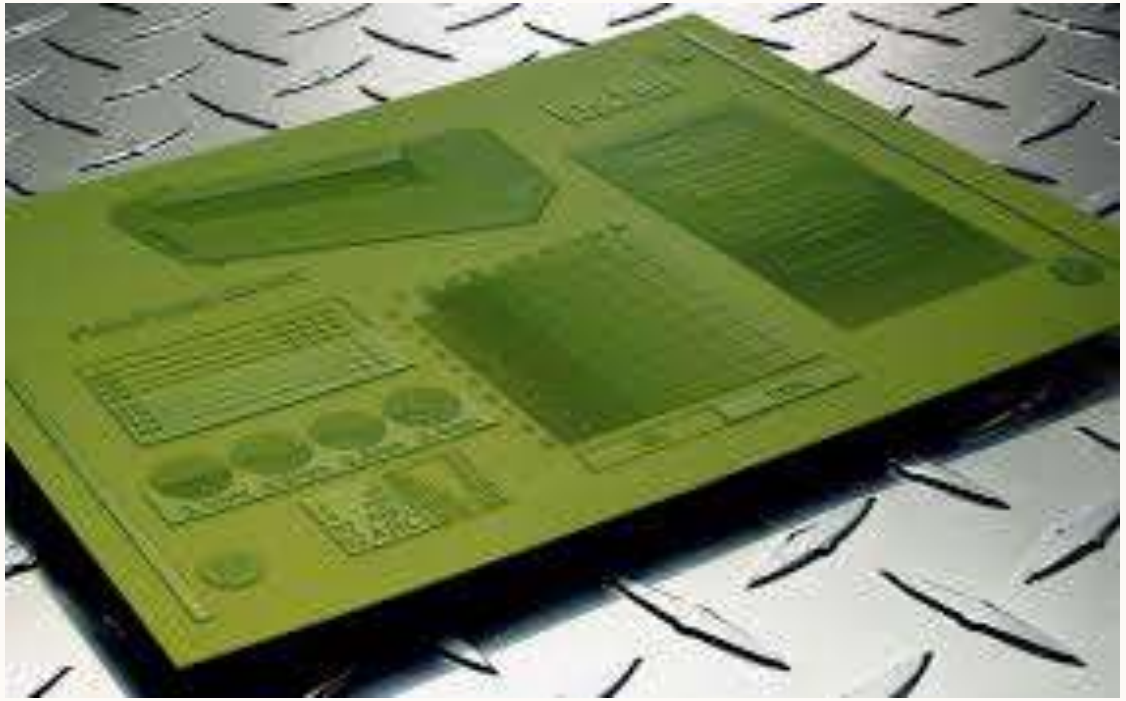


Double Layer CTP Plate Structure




Over coat layer
Photosensitive layer
Under coat layer
Aluminium base







B - POLYESTER PLATES

- Polyester plates or plastic plates are less expensive than metal plates and are a good alternative for printing quantities of 25,000 or less. These plates are used for printing small, color-coded prints with high quality. Technological advancements have helped eliminate plate expansion problems when printing large quantities, which previously caused expansion and inaccurate registration. Improved output device technology has also contributed to the success of polyester plates. Polyester sheets are available with a variety of sensitive coatings, making them compatible with a wide variety of lasers. Most commonly used sheets are exposed to a red or infrared laser.
- The most common coatings used for polyester sheets are silver halides, which are covered with a toner-based coating. When exposed to the laser, the printed areas fuse with heat.
- Slab thicknesses vary, ranging from 0.004 to 0.012 inches. Thicker sheets are used for large-volume printing, as the thickness makes the sheets more durable and resilient.

C - PAPER PLATES

Paper plates are made from cellulose and come in various thicknesses ranging from 0.004 to 0.008 inches.

These plates are not grained, but are chemically treated during manufacturing, so the non-printing areas retain a thin layer of wetting solution. These plates are used for printing small quantities in small presses, and they operate using toner-based technology for plate recording.

STAGES OF TRADITIONAL PLATE PREPARATION

- Photolithography
- Chemical treatment
- Plate piercing
- Edge bending

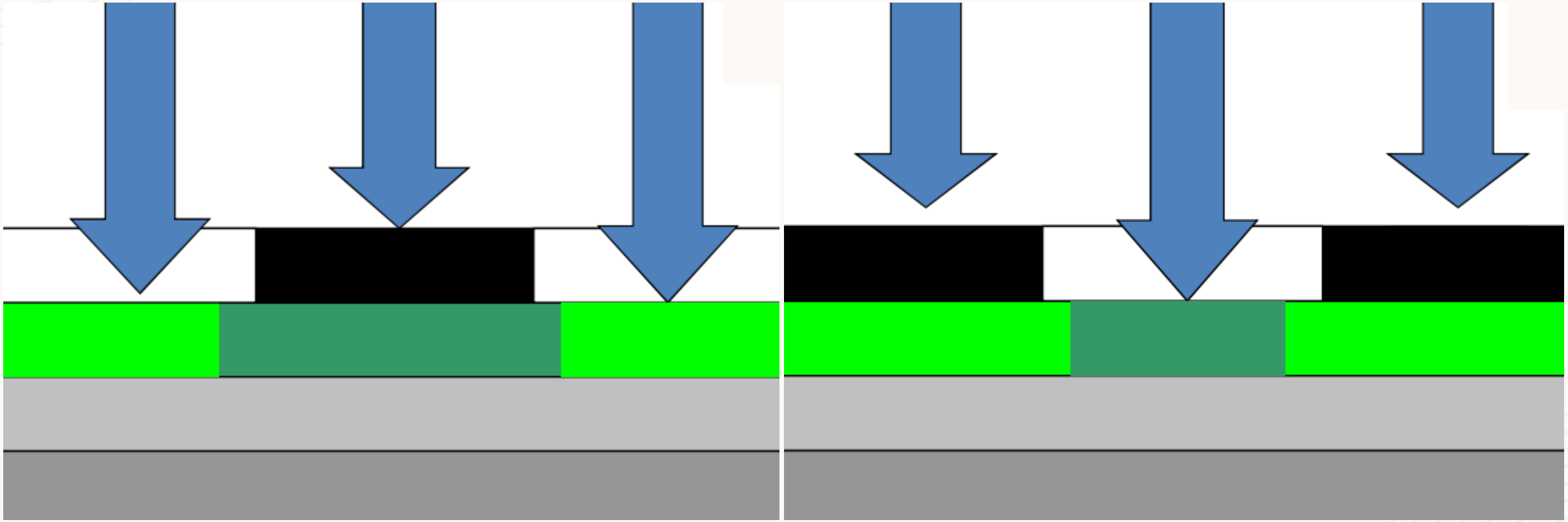




THE PHOTOLITHOGRAPHY STAGE:

- By placing the montage film in full contact with the printing plate, the correct position on the printing plate (taking into account the size of the pin),
- ensure the glass is clean and free of any dust or fingerprints, ensure the suction power, ensure the light source is strong, and determine the appropriate exposure time.
- This is accomplished using the photolithography unit, which varies in shape, size, and method for controlling the amount of light (per second or per unit).

Increasing/ Reducing exposure time for negative-performance material



Point Size

CHEMICAL PROCESSING STAGE

- Chemical processing is carried out through:

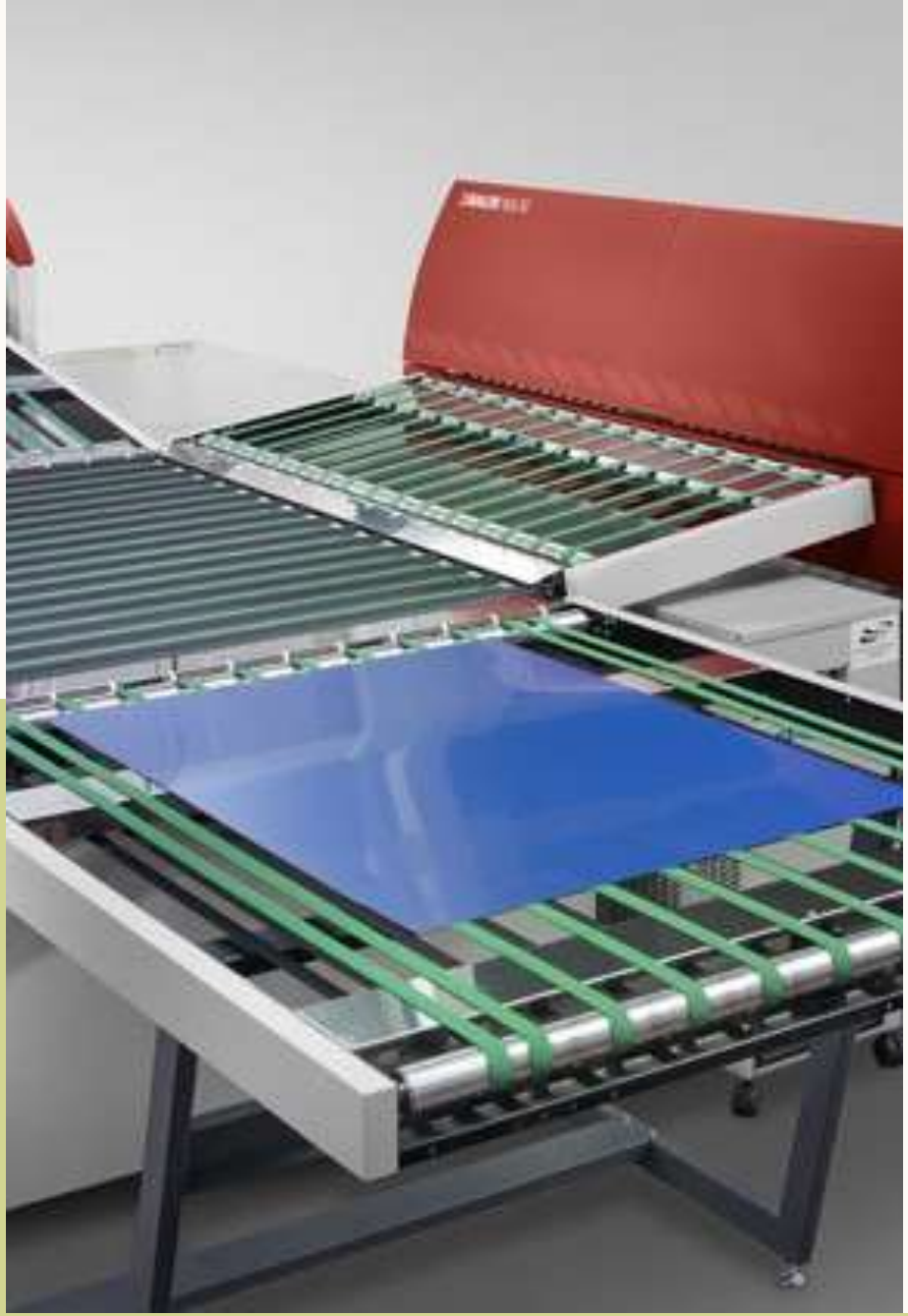
1. Manual Method

2. Automated Method

- Automated developing equipment is generally designed for the primary purpose of developing speed, with calibration for all stages of the development process.

This involves controlling the following:

- The temperature of the various developing unit.
- The supply ratio of the developing solution and washing water.
- Controlling the glue supply ratio and the required viscosity.



STAGES OF THE CHEMICAL PROCESSING PROCESS FOR PRINTING PLATES:

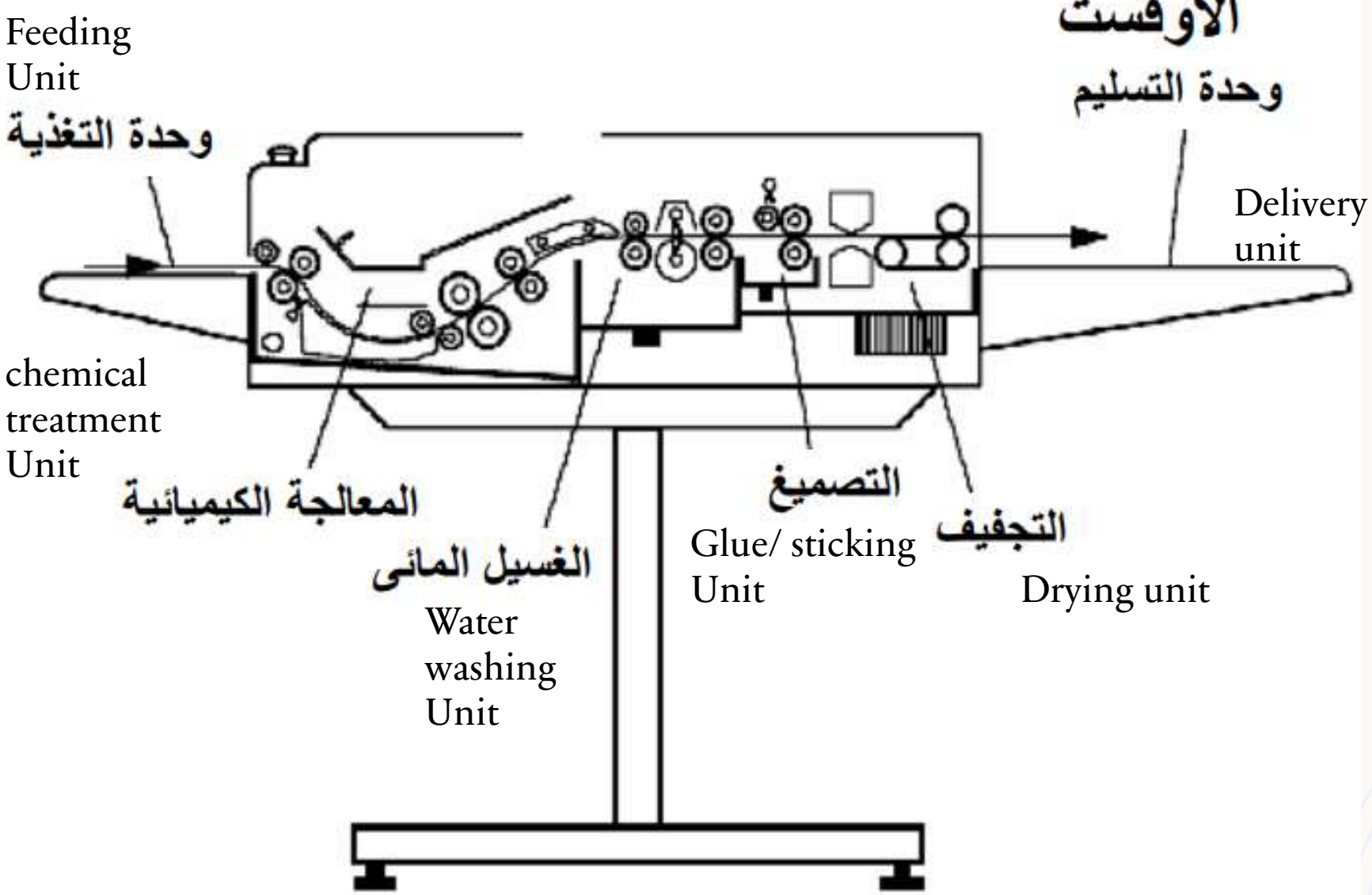
– Chemical development

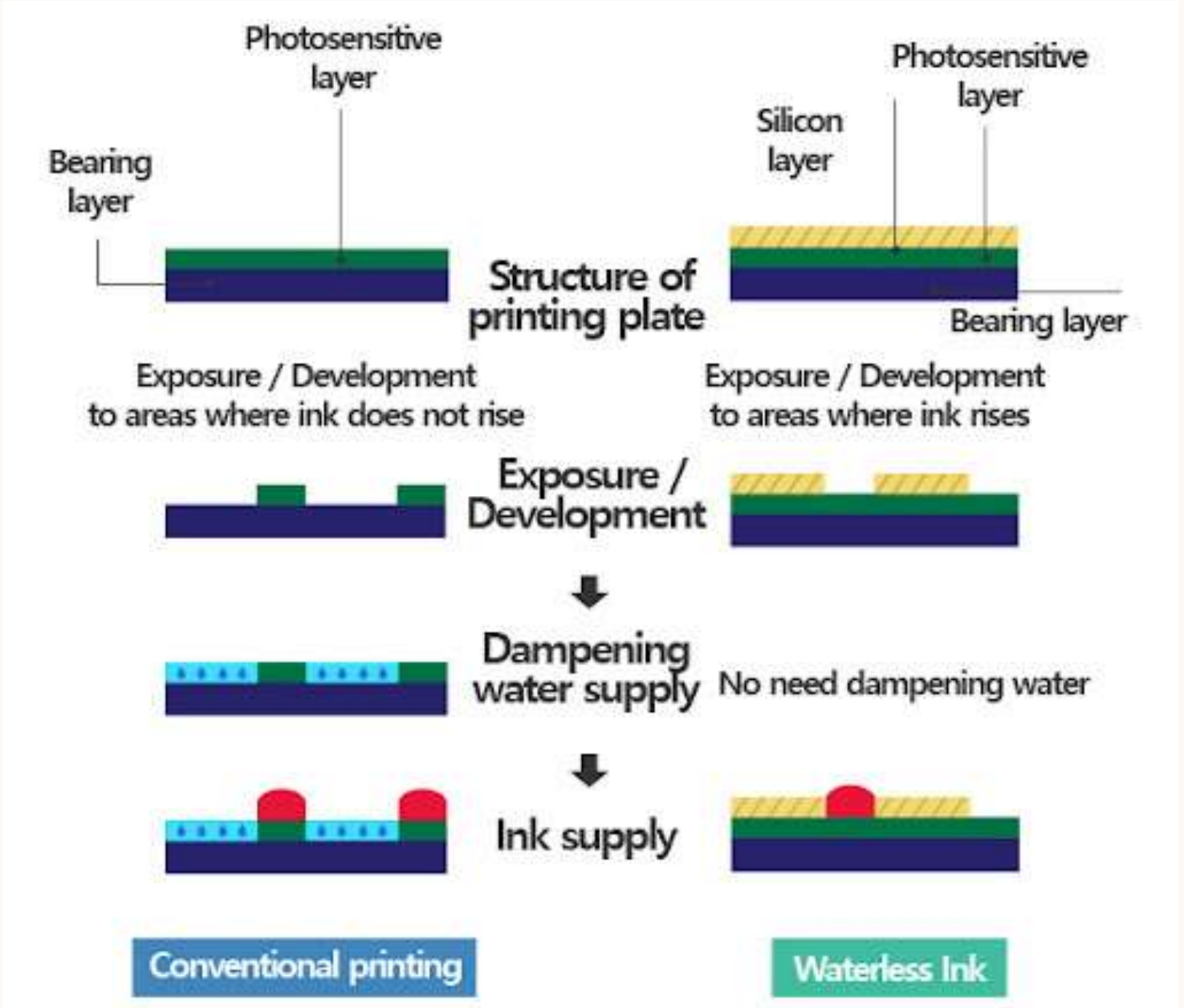
– Water washing

– Gluing

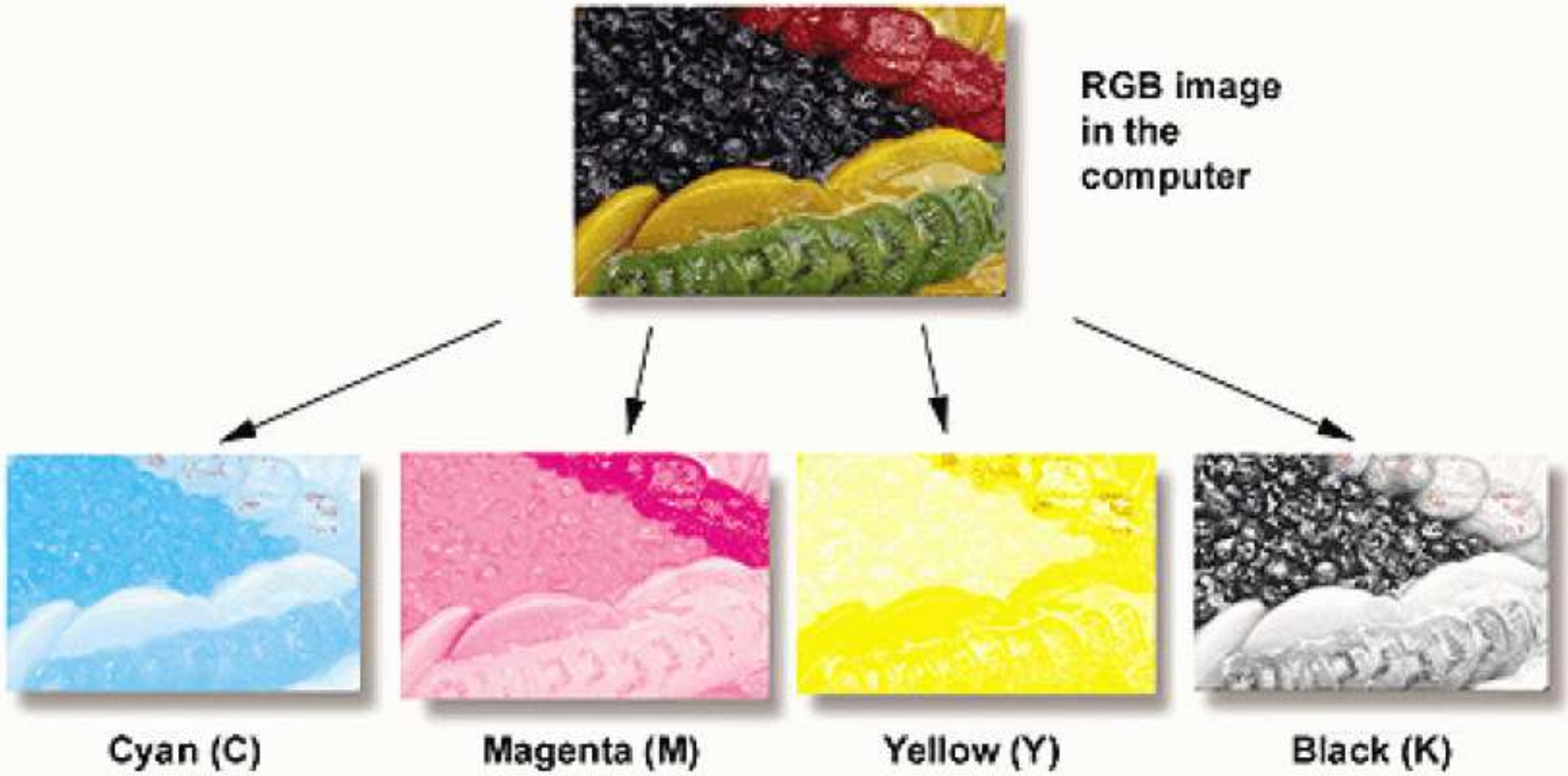
– Drying

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Photosensitive layer
Bearing layer
Structure of printing plate
Photosensitive layer
Silicon layer
Bearing layer
Exposure / Development to areas where ink does not rise
Exposure / Development to areas where ink rises
Exposure / Development
Dampening water supply
No need dampening water
Ink supply
Conventional printing
Waterless Ink

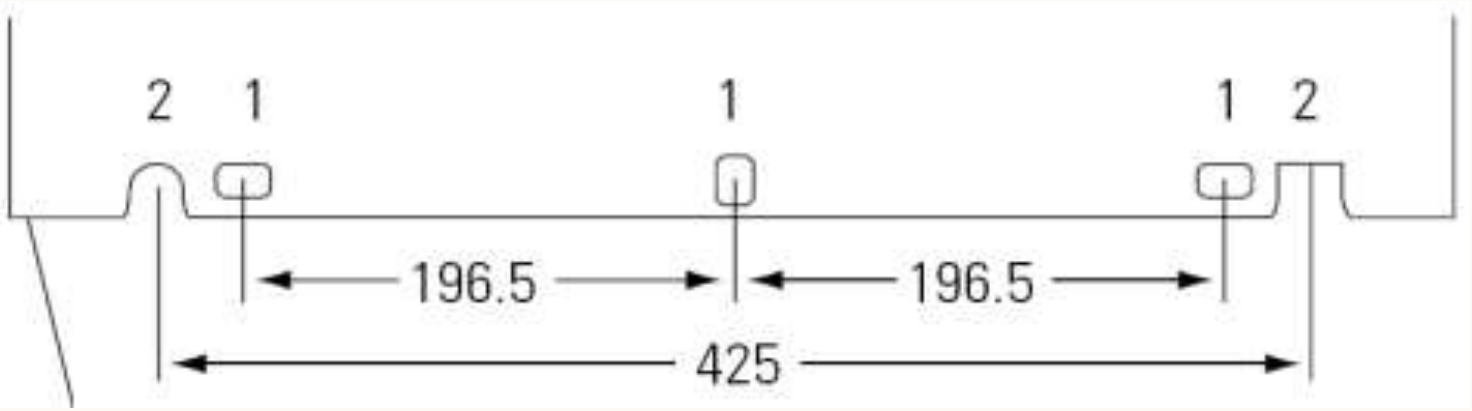

Stacker
Prosessor
UV-800FA
Autoloader


RGB image
in the
computer
Cyan (C)
Magenta (M)
Yellow (Y)
Black (K)


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PERFORATING THE PRINTING SURFACES

- The printing plate is perforated with specially designed holes, allowing for:
- Secure film placement without any movement
- Standardized color matching
- Ensuring the printing surface is correctly positioned on the printing surface cylinder

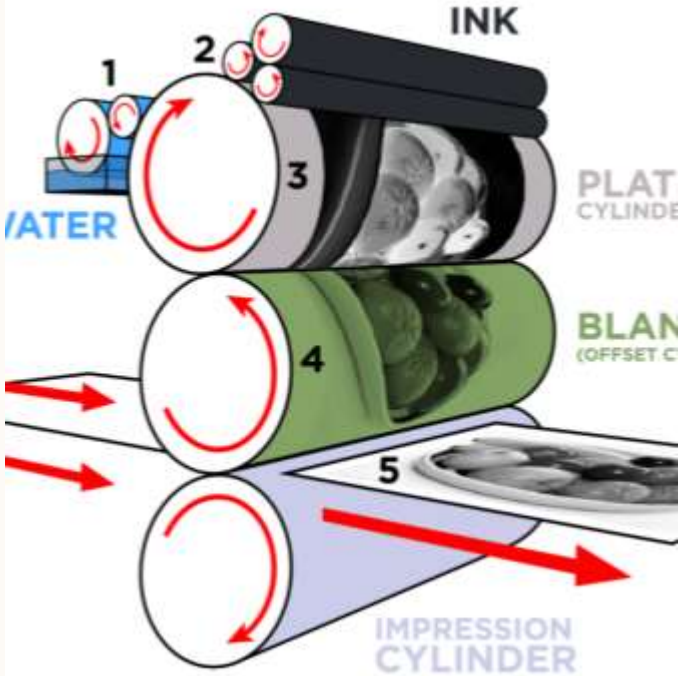


EDGE BENDING UNIT

- Bending the edges of the printing surface to allow it to be mounted on a cylinder.
- The printing surface is mounted on the printing machine by:
  - Attaching the printing surface to the designated place inside the machine through the holes in the printing plate and ensuring their presence with the pins in the machine
    - Two levers grip the edges of the printing surface and bend them.

SAFETY AND SECURITY IN LITHOGRAPHY SURFACE PREPARATION DEPARTMENTS:

- Ensure ventilation throughout the department.
- Wear gloves to avoid direct skin contact with the chemicals used.
- Ensure that none of the chemicals reach the human eye.
- Maintain the temperature between 18 and 22°C.
- A cabinet containing first aid supplies and equipment must be available.



QUESTIONS?